

Lab 04: DHCP Configuration

Duration: 1 hour | In-Lab Assessment

Tools: Cisco Packet Tracer

Objectives

By the end of this lab, students will be able to:

- Set up DHCP both via a dedicated server and via a Cisco router.
- Inspect DHCP bindings and troubleshoot IP lease assignments.

Lab Description

Dynamic Host Configuration Protocol (DHCP)

DHCP is a network management protocol used to automate the process of configuring devices on IP networks, thus allowing them to use network services such as DNS, NTP, and any communication protocol based on UDP or TCP. A DHCP server dynamically assigns an IP address and other network configuration parameters to each device on a network so it can communicate with other IP networks.

The DHCP server maintains a database of available IP addresses and configuration information. When it receives a request from a client, the DHCP server determines the network to which the DHCP client is connected, then allocates an IP address or prefix appropriate for the client from that network, and sends configuration information appropriate for that client.

The DHCP server and DHCP client must be connected to the same network link. In larger networks, each network link contains one or more DHCP relay agents. These DHCP relay agents receive messages from DHCP clients and forward them to DHCP servers. DHCP servers send responses back to the relay agent, and the relay agent then sends these responses to the DHCP.

DHCP servers typically grant IP addresses to clients for a limited interval called a lease. DHCP clients are responsible for renewing their IP address before that interval has expired and must stop using it once it has expired if they have not been able to renew it.

Configure DHCP (using server)

A router of device model 2811, two switches of device model 2960, two PCs, two laptops, and one server have been used in the sample topology shown in [Figure 1](#) for the DHCP configuration.

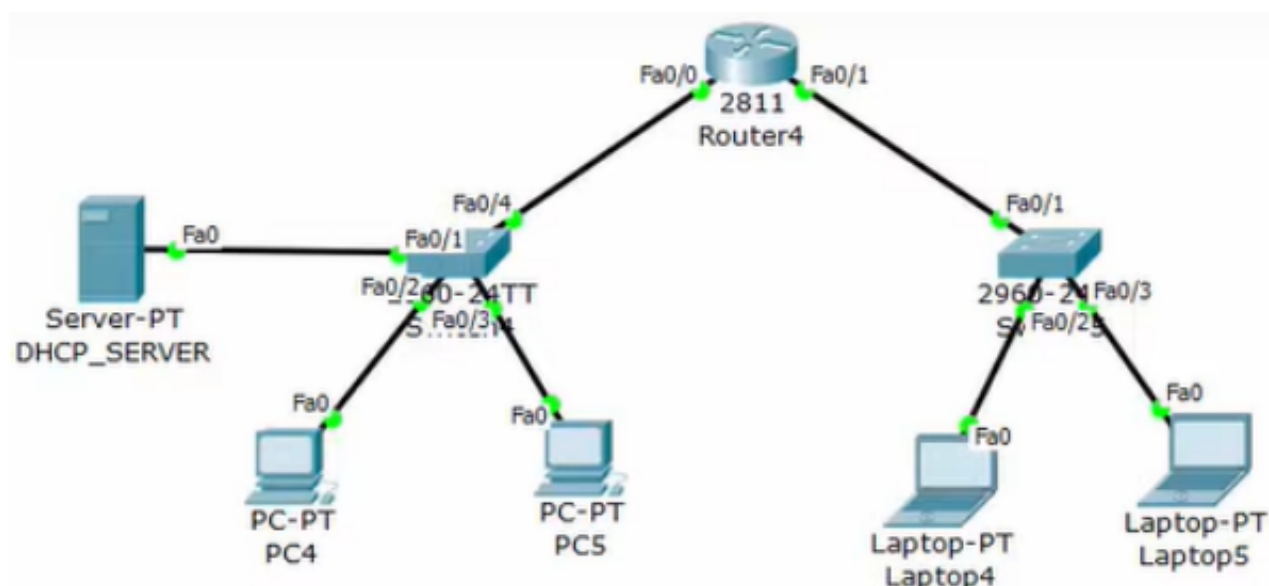


Figure 1: A sample network topology for the configuration of DHCP.

1. Configure DHCP Server

```

1 IP Address: 192.168.1.2
2 Default: 192.168.1.1

```

2. Make DHCP Pools

Go to Services and then DHCP, and change the following fields.

```

1 Pool Name: dotONEnetwork
2 Default: 192.168.1.1
3 Start IP: 192.168.1.3
4 Max Number: 20

```

Select **Add** to add the Address Pool to the server. Select **Save** to save the modification to the server.

To add another pool to the server, change the information as follows and then hit **Add** and **Save** buttons again.

```

1 Pool Name: dotTWOnetwork
2 Default: 192.168.2.1
3 Start IP: 192.168.2.2
4 Max Number: 20

```

Do not forget to turn on the DHCP server.

3. Configure R1 Interfaces

```

1 R1(config)#int fa0/0
2 R1(config-if)# ip address 192.168.1.1 255.255.255.0
3 R1(config-if)# ip helper-address 192.168.1.2
4 R1(config-if)#no shutdown
5 R1(config-if)#exit
6 R1(config)#int fa0/1
7 R1(config-if)# ip address 192.168.2.1 255.255.255.0

```

```
8 R1(config-if)# ip helper-address 192.168.1.2
9 R1(config-if)#no shutdown
10 R1(config-if)# end
11 R1# copy running-config startup-config
```

The **ip helper-address** command should be applied on the inbound Layer 3 interface (such as a router interface or VLAN interface) that directly receives the broadcast traffic from the client devices.

4. Configure all the PCs

Just click DHCP and the server will do the rest.

5. Verify

Ping PC1 from PC0

Configure DHCP (using router)

The exact network topology with the same device models, as shown in [Figure 1](#), has been used for this configuration.

Configure the DHCP server in the router instead of a server.

```
1 R1(config)# ip dhcp pool dotONEnetwork
2 R1(dhcp-config)# default-router 192.168.1.1
3 R1(dhcp-config)# network 192.168.1.0 255.255.255.0
4 R1(dhcp-config)# exit
5
6 R1(config)# ip dhcp pool dotTWOnetwork
7 R1(dhcp-config)# default-router 192.168.2.1
8 R1(dhcp-config)# network 192.168.2.0 255.255.255.0
9 R1(dhcp-config)# exit
```

The rest of the configuration, i.e., Configure R1 Interfaces, Configure all the PCs, and Verify are same as the section .

Use **show ip dhcp binding** inside the router to see the status of the configured DHCP.

Student ID:

Lab 04 Task

Your company, Make Abdullahpur Great Again (MAGA), is expanding its headquarters and requires a new network design to support various departments. The MAGA headquarters consists of the following departments across multiple floors: Engineering - Floor 1, Marketing - Floor 2, Sales - Floor 3, Administration - Floor 4, Management - Floor 5

Suppose they all share the prefix **192.(i).(i+3) % 10**. Below are the required devices for each of the departments:

$E(i) = (5i + 3) \bmod 41 + 5$	(Engineering)
$M(i) = (7i + 11) \bmod 26 + 5$	(Marketing)
$S(i) = (3i + 7) \bmod 36 + 5$	(Sales)
$A(i) = (11i + 17) \bmod 18 + 5$	(Administration)
$G(i) = (13i + 1) \bmod 9 + 5$	(Management)

where i is the last two digits your student ID

Implement and configure the topology, the devices should get their IP automatically from the DHCP server. Now for Engineering, Marketing and Sales the devices should use a separate DHCP server for automatic IP assignment & for Admin and Management they should use a router as a DHCP server. For each network you need to add minimum 4 end devices.

Lab 04 Grading Criteria

- 1. Correct Topology & Router Configuration (15 pts)**
 - Correct topology (5 pts)
 - Correct router configuration (10 pts)
- 2. DHCP Configuration (20 pts)**
 - Correct DHCP server configuration (10 pts)
 - Correct router as a DHCP server configuration (10 pts)
- 3. Routing (5 pts)**
 - Correct RIP configuration (5 pts)
- 4. Total: 40 pts**